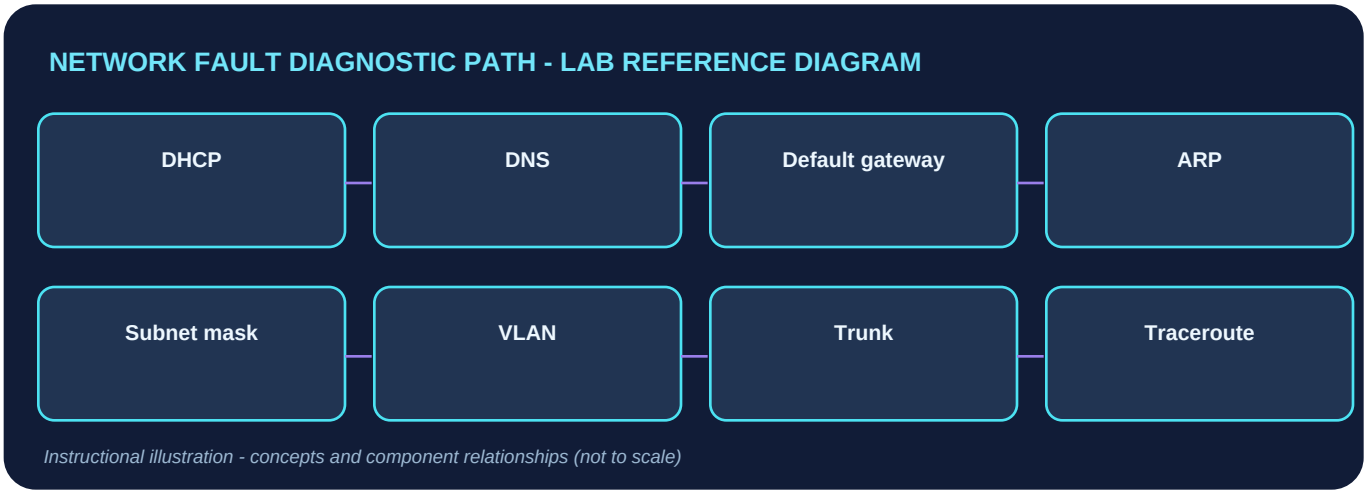
5.3 Network Troubleshooting

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Resolve DHCP, DNS, cabling, gateway and VLAN faults.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Check physical link and interface state.
2. Review IP, subnet mask, gateway and DHCP lease.
3. Ping gateway then test DNS name resolution.
4. Inspect switch VLAN and routed path if needed.

Troubleshooting and safe practice

• Ping IP but not name: investigate DNS.

• Wrong VLAN: inspect access port and trunk tagging.

5.3 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
DHCP	Service that automatically leases IP settings to clients.
DNS	Service that translates names into IP addresses.
Default gateway	Router address used for destinations outside the local network.
ARP	Protocol used to learn Layer 2 addresses for local IPv4 peers.
Subnet mask	Identifies network and host bits in an IPv4 address.
VLAN	Logical Layer 2 segmentation of switch ports and networks.
Trunk	Switch link carrying traffic for multiple VLANs using tagging.
Traceroute	A key term in network fault diagnostic path; identify its purpose, where it is used, and how to verify it.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____